

Haseeb Ashfaq

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EXPERIENCE

Google

Software Engineer, Ph.D.

June 2026 – Present

California, USA

- Part of the AI and Infrastructure organization, working on reliable low-latency networking and ML telemetry
- Currently working on low-jitter host networking through HW/SW co-designed algorithms on SmartNICs

Google

Software Engineering Intern, Ph.D.

May 2025 – August 2025

California, USA

- Part of the AI and Infrastructure organization; worked on improving the robustness of ML infrastructure
- Developed a tool, *hstprof*, for profiling GPU/TPU workloads using high-speed network telemetry
- *hstprof* enabled a fine-grained view of the network traffic of ML jobs, reporting utilization at a 100-microsecond timescale
- Used *hstprof* to analyze Gemini training clusters to investigate several regressions, including packet drops and high latencies
- Found an imbalance of packet queues across the memory banks of switches, which caused performance degradation

Nokia Bell Labs

Networking Research Intern

June 2023 – August 2023

New Jersey, USA

- Developed (in C++, Unix) a streaming service for AR/VR content for heterogeneous networks
- Implemented a resource-efficient transcoding mechanism for volumetric videos that achieved 75% CPU savings
- Developed an encoder/decoder for point cloud data that can tolerate packet losses in the network, which enabled utilizing an unreliable transport protocol (UDP) instead of TCP for point cloud streaming

Networks and Distributed Systems Group, NYU

Graduate Research Assistant

Sep. 2025 – May 2026

New York, USA

- Designed a special priority queue, LOQ, for cloud-hosted financial exchanges, that enhances a matching engine's throughput by up to 150% and lowers latency by 90% while retaining fairness among market participants
- Designed a cloud multicast service, in C++, for market data, delivering lower latency and greater scalability than AWS

PosterMyWall

Software Engineer (Full Time)

June 2020 – August 2021

Lahore, PK

- Developed, in PHP and JS, an access control system for internal tools of the company
- Set up a CI/CD pipeline along with testing infrastructure using TeamCity and AWS
- Automated AWS-hosted development infrastructure, shortening the testing cycle time by more than 50%
- Secured the product website by eliminating critical vulnerabilities (XSS, CSRF, IDOR) and did backend development

EDUCATION

PhD and MS, Computer Science

New York University, New York, USA

Sep. 2021 – May 2026

GPA: 4.0/4.0

Research Interests: Distributed Systems, Networks, Cloud Computing, Financial Technologies, AI Infrastructure

Bachelor of Science, Computer Science

Lahore University of Management Sciences, Lahore, Pakistan

Sep. 2016 – May 2020

GPA: 3.7/4.0

Courses: Algorithms, Data Structures, Distributed Systems, Computer Networks, Machine Learning

PROJECTS

Design and Implementation of a Low-Latency and Scalable Cloud Financial Exchange

- Implemented a low-latency market data multicast, achieving a 1-microsecond latency difference across receivers
- Utilized DPDK kernel bypass and zero-copy packet replication techniques, implemented in C++
- Utilized eBPF/XDP and eBPF/TC for efficient packet processing when using the Linux kernel

KV Cache-Optimized Enterprise LLM Coding Agent

- Designed a coding agent that maximized KV cache usage across members of a team, reducing inference cost
- As several engineers of a team share base source code, the coding agent constructs prompts as a combination of base source files and per-engineer patches, along with several optimizations to enhance the common prefix across prompts
- Reduced LLM prefill computation substantially through ~85% resultant common prompt prefixes

PUBLICATIONS

Beyond Lamport, Towards Probabilistic Fair Ordering

[ACM HotNets'25](#)

Network Support For Scalable and High-Performance Cloud Exchanges

[ACM SIGCOMM '25](#), Cited by [Jane Street](#), [Google](#), and [Microsoft Research](#)

ParserHawk: Hardware-aware parser generator using program synthesis

[ACM SIGCOMM '25](#)

Towards Efficient Transmission of 3D Point Clouds Via Adaptive Encoding and QUIC

[Emerging Multimedia Systems \(EMS\)'25](#), a SIGCOMM workshop

Codesign of Tensors' Encoding And Transcoding

[Networks for AI Computing \(NAIC\)'25](#), a SIGCOMM workshop

Patent: A Method To Enable Fast Transmission And Processing Of 3D Telepresence Data Encoded As Octrees

Approved by Nokia's internal board, in [submission to USPTO](#), received a monetary award from Nokia Bell Labs

QuEST: Fast, Expressive, and Cheap Analytics for Distributed Traces Using Cloud Storage

[CloudDB](#), a VLDB workshop

INVITED TALKS

Open Compute Project, OCPTAP: Network Support For Scalable And High Performance Cloud Exchanges. 10/22/2025

Google: How to build an ultra-fast and scalable financial exchange on the public cloud? 12/03/2024

Rutgers University: Network support for cloud-hosted financial exchanges. 10/30/2024

AWARDS, FELLOWSHIPS, AND SERVICES

Outstanding Research Award

Granted by Nokia Bell Labs during the Global Student Program 2023

National Science Foundation (NSF) Grant

Funds for traveling to ACM SIGCOMM 2024 in Sydney, Australia

Reviewer for academic research articles

Served as a reviewer for research articles submitted to ACM JCSS and IEEE Transactions on Cloud Computing

MacCracken Fellowship

Granted by New York University for a Ph.D. in Computer Science

SKILLS

C/C++, Python, PHP, SQL, Go, JavaScript, React/React Native, Rust, AWS, Debugging, Testing, DPDK, eBPF, Linux, Kubernetes, Docker, Istio, Microservices, Congestion Signalling (CSig), High-Speed Network Telemetry, System Design, KV Cache Optimization

MISC.

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Available for full-time: Not actively looking